Substitution and Formulae

Name _____ **Date** _____ No calculator Version 1

KEY WORDS substitute · expression · formula · value · subject · rearrange

A Substituting into expressions ●○○

ANSWER ON THE SHEET

Remember: Replace each letter with its number. Remember that $3a$ means $3 \times a$ and ab means $a \times b$.

1. Find the value of each expression when $a = 4$, $b = 3$ and $c = 6$.

a) $a + b$ = _____ **b)** $3a$ = _____ **c)** $c - b$ = _____ **d)** ab = _____

e) $2a + c$ = _____ **f)** $5b - a$ = _____ **g)** a^2 = _____ **h)** $bc - a$ = _____

i) $\frac{c}{b}$ = _____ **j)** $2(a + b)$ = _____ **k)** abc = _____ **l)** $c^2 - a^2$ = _____

m) $3a + 2b$ = _____ **n)** $\frac{a + c}{2}$ = _____ **o)** $10 - b$ = _____ **p)** $4(c - a)$ = _____

B Substituting negative numbers ●●○

WORK IN YOUR BOOK

Watch out: Put brackets around a negative number when you substitute it: $(-3)^2 = 9$, not -9.

2. Find the value of each expression when $x = -3$, $y = 2$ and $z = -1$.

a) $x + y$ **b)** $y - x$ **c)** xy **d)** x^2

e) $2x + y$ **f)** xz **g)** $y - z$ **h)** $3z + 5$

i) $x + y + z$ **j)** $z^2 + y$ **k)** xyz **l)** $2y - 3x$

C Using formulae ●●○

WORK IN YOUR BOOK

3. Substitute into each formula.

a) Perimeter of a rectangle: $P = 2l + 2w$. Find P when $l = 9$ and $w = 4$.

b) Area of a triangle: $A = \frac{1}{2}bh$. Find A when $b = 10$ and $h = 6$.

c) Distance: $d = st$. Find d when $s = 60$ and $t = 2.5$.

d) A mechanic charges C dollars for h hours: $C = 25 + 40h$. Find C when and $h = 3$.

e) $v = u + at$. Find v when $u = 2$, $a = 5$ and $t = 3$.

f) $y = 3x - 7$. Find y when and $x = 4$.

g) Area of a square: $A = l^2$. Find A when and $l = 9$.

h) Minutes to cook a roast of n kilograms: $T = 20n + 15$. Find T when and $n = 6$.

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D Rearranging: one step ●●○ — WORK IN YOUR BOOK

> **Remember:** To make a letter the subject, undo what has been done to it. Do the same to both sides.

4. Make the letter in square brackets the subject of the formula.

a) $y = x + 4 \quad [x]$

b) $y = x - 9 \quad [x]$

c) $y = 5x \quad [x]$

d) $y = \frac{x}{3} \quad [x]$

e) $A = lw \quad [l]$

f) $d = st \quad [t]$

g) $p = q + r \quad [q]$

h) $m = \frac{n}{k} \quad [n]$

i) $P = 4s \quad [s]$

j) $C = \pi d \quad [d]$

k) $t = w - 12 \quad [w]$

l) $V = Ah \quad [h]$

E Rearranging: two steps ●●● — WORK IN YOUR BOOK

> **Remember:** Undo adding and subtracting first. Then undo multiplying and dividing.

5. Make the letter in square brackets the subject of the formula.

a) $y = 2x + 3 \quad [x]$

b) $y = 5x - 1 \quad [x]$

c) $y = mx + c \quad [x]$

d) $v = u + at \quad [t]$

e) $y = \frac{x}{2} + 5 \quad [x]$

f) $P = 2l + 2w \quad [l]$

g) $C = 25 + 40h \quad [h]$

h) $y = 3(x + 2) \quad [x]$

i) $A = \frac{1}{2}bh \quad [b]$

★ Extension — FOR EARLY FINISHERS

6. These formulae have squares and square roots. Make the letter in square brackets the subject.

a) $A = s^2 \quad [s]$

b) $y = x^2 + 1 \quad [x]$

c) $y = \sqrt{x} \quad [x]$

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Substitution and Formulae

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS substitute · expression · formula · value · subject · rearrange

A Substituting into expressions ●○○ — ANSWER ON THE SHEET

Remember: Replace each letter with its number. Remember that $3a$ means $3 \times a$ and ab means $a \times b$.

1. Find the value of each expression when $a = 5$, $b = 2$ and $c = 8$.

a) $a + b$ = _____ **b)** $4a$ = _____ **c)** $c - b$ = _____ **d)** ab = _____

e) $2a + c$ = _____ **f)** $6b - a$ = _____ **g)** a^2 = _____ **h)** $bc - a$ = _____

i) $\frac{c}{b}$ = _____ **j)** $3(a + b)$ = _____ **k)** abc = _____ **l)** $c^2 - a^2$ = _____

m) $2a + 3b$ = _____ **n)** $\frac{a + c + 1}{2}$ = _____ **o)** $10 - b$ = _____ **p)** $5(c - a)$ = _____

B Substituting negative numbers ●●○ — WORK IN YOUR BOOK

Watch out: Put brackets around a negative number when you substitute it: $(-3)^2 = 9$, not -9.

2. Find the value of each expression when $x = -2$, $y = 5$ and $z = -4$.

a) $x + y$ **b)** $y - x$ **c)** xy **d)** x^2

e) $3x + y$ **f)** xz **g)** $y - z$ **h)** $2z + 9$

i) $x + y + z$ **j)** $z^2 - y$ **k)** xyz **l)** $2y - 3x$

C Using formulae ●●○ — WORK IN YOUR BOOK

3. Substitute into each formula.

a) Perimeter of a rectangle: $P = 2l + 2w$. Find P when $l = 8$ and $w = 5$.

b) Area of a triangle: $A = \frac{1}{2}bh$. Find A when $b = 12$ and $h = 5$.

c) Distance: $d = st$. Find d when $s = 80$ and $t = 1.5$.

d) A mechanic charges C dollars for h hours: $C = 30 + 45h$. Find C when and $h = 2$.

e) $v = u + at$. Find v when $u = 3$, $a = 4$ and $t = 5$.

f) $y = 4x - 9$. Find y when and $x = 3$.

g) Area of a square: $A = l^2$. Find A when and $l = 7$.

h) Minutes to cook a roast of n kilograms: $T = 25n + 10$. Find T when and $n = 4$.

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D Rearranging: one step ●●○ — WORK IN YOUR BOOK

> **Remember:** To make a letter the subject, undo what has been done to it. Do the same to both sides.

4. Make the letter in square brackets the subject of the formula.

a) $y = x + 7 \quad [x]$

b) $y = x - 2 \quad [x]$

c) $y = 6x \quad [x]$

d) $y = \frac{x}{4} \quad [x]$

e) $A = bh \quad [b]$

f) $d = st \quad [s]$

g) $p = q + r \quad [r]$

h) $m = \frac{n}{k} \quad [n]$

i) $P = 3s \quad [s]$

j) $C = \pi d \quad [d]$

k) $t = w - 15 \quad [w]$

l) $V = Ah \quad [A]$

E Rearranging: two steps ●●● — WORK IN YOUR BOOK

> **Remember:** Undo adding and subtracting first. Then undo multiplying and dividing.

5. Make the letter in square brackets the subject of the formula.

a) $y = 3x + 2 \quad [x]$

b) $y = 4x - 5 \quad [x]$

c) $y = mx + c \quad [x]$

d) $v = u + at \quad [a]$

e) $y = \frac{x}{3} + 4 \quad [x]$

f) $P = 2l + 2w \quad [w]$

g) $C = 30 + 45h \quad [h]$

h) $y = 4(x + 1) \quad [x]$

i) $A = \frac{1}{2}bh \quad [h]$

★ Extension — FOR EARLY FINISHERS

6. These formulae have squares and square roots. Make the letter in square brackets the subject.

a) $A = l^2 \quad [l]$

b) $y = x^2 - 3 \quad [x]$

c) $p = \sqrt{q} \quad [q]$

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